Autonomous Vehicle Trajectory Prediction

Behavioral AI & Temporal Modeling

Proposal & Team Details

Proposal Title:

Social Transformer for Multi-Modal Pedestrian & Cyclist Trajectory Prediction

Team Name: guava juice

Team Lead: Robin Punnen Alex, robinalexpnn@gmail.com, MIT Manipal

Challenge Track: AI and Computer Vision

Team Members, Affiliation & Expertise

Member 1 (Team Lead): Robin Alex – MIT Manipal – Primary Expertise(Deep Learning)

Member 2 : Sathvik Reddy – MIT Manipal – Expertise (Computer Vision / Deep Learning)

Member 3 : Aaryan Salunke – MIT Manipal – Expertise(Deep Learning)

Understanding of the Problem Statement

Predicting pedestrian and cyclist trajectories in L4 urban environments
In an L4 urban environment, reacting to where an agent is isn't enough — the vehicle must predict where they will be.

- Key objective: Predict future (x, y) coordinates (next 3 seconds) from 2 seconds of past motion
- Type of task: Temporal sequence prediction / multi-agent trajectory forecasting
- Real-world impact: Enables safe autonomous driving decisions in crowded urban environments

Proposed Solution – High Level Architecture

Multi-Modal Transformer with Social Pooling
End-to-end trajectory prediction pipeline:
nuScenes Input Data
→ Preprocessing (coordinate offsets + neighbor extraction)
→ Dual Embedding (target trajectory + neighbor trajectories)
→ Social Pooling (max-pool over neighbors)
→ Transformer Encoder (4 layers, 4 heads)
→ Multi-Modal Output (3 future paths + confidence scores)
Architecture outputs 3 possible future paths with probability scores for each

Methodology

Step-by-step workflow for trajectory prediction:
Dataset collection and preparation:
nuScenes v1.0-mini dataset, filtered for pedestrians & cyclists
Preprocessing: Anchor trajectories to current position, compute step-by-step coordinate offsets (deltas), extract up to 5 nearest neighbors
Model training: 70 epochs, batch size 32, AdamW optimizer (lr=2e-3) with cosine annealing LR schedule
Validation: Track ADE and FDE each epoch; apply gradient clipping (max norm=1.0)
Inference: Multi-modal predictions with mode probability scoring and visualization

Techniques & Algorithms

Core models and algorithms implemented:
Two architectures explored:

- Social LSTM (Baseline): LSTM encoder-decoder with spatial grid for neighbor encoding
- Multi-Modal Transformer (Main): TransformerEncoder (4 layers, 4 heads, d_model=128) with social pooling
- Offset-based coordinates: Predict step-by-step deltas for training stability
- Social Pooling: Max-pool over neighbor embeddings for social context
- Multi-modal output: 3 predicted modes with softmax probability scores

Dataset & Training Strategy

Dataset: nuScenes (v1.0-mini)

- Categories: human.pedestrian.adult, vehicle.bicycle
- Past: 2 seconds at 2Hz = 4 frames; Future: 3 seconds at 2Hz = 6 frames
- Up to 5 nearest neighbors extracted per agent per sample
- Optimizer: AdamW (lr=2e-3, weight_decay=1e-4) — 70 epochs
- Scheduler: Cosine Annealing LR
- Gradient clipping (max norm=1.0) to prevent exploding gradients
- Hardware: Google Colab T4 GPU

Real-Time Processing & Performance

System performance and robustness:

- Single forward pass per agent for fast inference
- Lightweight model: d_model=128, 4 Transformer layers
- Offset-based inputs normalize absolute position → robust across map locations
- Zero-padding for missing neighbors → handles variable crowd density

Robustness considerations:

- Stable training due to delta/offset coordinate representation
- Social context naturally handles crowded pedestrian scenarios
- Multi-modal outputs handle inherent uncertainty in future trajectories

Evaluation Metrics

Performance measured using displacement-based metrics:
Primary metrics:

- ADE (Average Displacement Error): Mean Euclidean distance over all future timesteps
- FDE (Final Displacement Error): Distance at the final predicted timestep
- minADE: Best of 3 predicted modes — rewards multi-modal diversity
- Cross-entropy loss for mode probability calibration

Training loss = minADE + cross-entropy (joint optimization of accuracy and mode selection)

GitHub Submission Requirement

Repository: github.com/robinpnalex/guava-juice (publicly accessible)

- Full project code: Social lstm.py, transformermodel.py, transformermodel.ipynb
- Well-documented README.md with all required sections

README covers:

- Project overview
- Model architecture (Social LSTM + Transformer)
- Dataset used (nuScenes)
- Setup & installation instructions
- How to run the code (local & Google Colab)
- Example outputs / prediction visualizations (3-mode plots)

Repository is publicly accessible during evaluation.

Declaration

We confirm that the proposed work is original and developed by the team

- All team members agree to the rules and evaluation process